Light in the Dark — Game Design Document v1

Game Development Fundamentals · Assignment 8–9 · Solo developer bl00dtearz (all roles: programming, design, art, UI, release) · github.com/bl00dtearz/light_in_the_dark

1. Title + pitch

Light in the Dark. A small lantern spirit crosses a pitch-black level where traps flash only for a split second — remember them, then walk through the dark.

2. Genre · platform

Troll platformer (hidden traps, tricks, instant failure). Android first, iOS second. Landscape only, reference resolution 1920 × 1080.

3. 2D or 3D

2D. Unity 6.3 LTS, URP 2D Renderer, Light 2D, pixel-art sprites (32 × 32 px, Point filter).

4. Core loop

Watch the flash → remember the traps → move through the dark → collect light orbs and reach the door (die → restart and try again). The fun is memory and surprise: every attempt the player knows a little more about the level.

5. Controls (on-screen, uGUI + Input System On-Screen Buttons)

Button	Position	Action
◀ Left / ▶ Right	bottom-left, 200 × 200 px each	walk left / right
JUMP	bottom-right, 240 × 240 px	jump (only on the ground)
II Pause	top-right, 120 × 120 px	pause: Resume, Restart, Menu

In the Editor the same actions work on the keyboard: A / D (or arrows) and Space.

6. Content (this assignment)

- 1 short level (60–90 s), one hero. **1 life:** one touch of a trap = death, the level restarts and the score resets.
- 8 light orbs, +10 score each. The exit door opens (Closed → Open) only when all 8 are collected.
- 2 trap types: **flashing spikes** (invisible, lit for 0.2 s every 3 s) and a **fake floor** (falls 0.3 s after you touch it); bottomless pits.
- 3 screens: Main Menu, HUD, Pause.

7. Art and sound

- **Style:** almost black world; the only light is the hero's lantern (Point Light 2D), the orbs, the door and the short trap flashes. Dark blue-grey stone + warm orange light, one world.
- **Sprites:** original pixel art made for this project (hero, stone tiles, spikes, orb, door, UI buttons, hearts). License: own work. Grid By Cell Size, Filter Point, Compression None.
- **Font:** TextMeshPro default, Liberation Sans (SIL Open Font License 1.1).
- **Sound (week 12):** ambient music loop + jump, orb, death, door, click — Kenney audio packs (CC0) or Freesound (CC0 only).

8. MVP (must be in the week-15 build)

- Main Menu, HUD and Pause working on 16:9, notch phones and 4:3 tablets.
- Hero with Idle / Walk / Jump animations and the light circle.
- **4 levels** (5th if time allows), each adding a trick: 1 – flashing spikes, fake floor, pits · 2 – spikes that flash only when you are near · 3 – the shadow enemy · 4 – a fake exit door, the real one is hidden · 5 (optional) – everything, and the light circle shrinks over time.
- Level select (a level unlocks after the previous one), 1 enemy type, saving (orb total, levels unlocked), sound, visual effects.

9. Cut list (we will NOT make these)

- Skin shop (orbs are saved from week 11, so a shop could be added after the course).
- More than 5 levels, a level editor · extra lives, checkpoints, hints · online leaderboards, achievements, multiplayer · cutscenes, a second character.

10. Plan, weeks 9–13

Week	Ready
9	GDD, GitHub repo, sprites, hero with Animator, Level 1 (traps, orbs, door), Main Menu / HUD / Pause with SafeArea
10	Enemy AI: a shadow creature that patrols the dark and chases the hero when he is near
11	Levels 2 and 3, level select, saving the orb total and unlocked levels (PlayerPrefs)
12	Sound: music loop and effects (jump, orb, death, door, buttons)
13	Visual effects (orb sparkles, death burst, lantern flicker), Level 4
14–15	Level 5 (optional), balance, release build, store page